Chapter 7. Machine Learning
Topic 7.1 Introduction

- Learning: Improving performance through careful study of experiences.
- Why learning:
  - Designers cannot anticipate all possible situations that the agent might find itself in
  - Designers cannot anticipate all changes over time
  - Designers may have no idea how to program a solution
- Inductive versus Deductive Learning
- Empirical, statistical, and example-based learning
- Explanation based learning: general rule from individual observation; Derivative(u^2, u) = 2u → Derivative(x^2, x) = 2x
- Types of learning based on feedback types:

  ** Unsupervised, for example, partitioning of objects without given labeled examples;
  ** Reinforcement, for example, fixing rules through rewards and punishments;
  ** Supervised, for example, discovering rules using labeled training dataset; (commonly used)

  *** Semi-supervised, Hypotheses verification, Generalization, etc.
- Learning models in practice: Linear regression, nonlinear regression, decision tree, neural network, naïve Bayes, clustering, support vector machines, etc.
- Learning by ensembles of models

The task of supervised learning is this:
Given a **training set** of N example input-output pairs
$(x_1, y_1), (x_2, y_2), \ldots, (x_N, y_N)$,
where each y_i was generated by an unknown function y=f(x),
discover a function h that approximates the function f.
x can be numeric or non-numeric; h is a hypothesis from a hypothesis space H.

Topic 7.2 Learning Decision Trees
- Form of inductive leaning
- A kind of Supervised learning

Example:

Decision Tree Induction for the concepts 'positive', that is, 'Buys a computer' and 'negative', that is, 'Does not buy a computer', from a set of Training Data (samples / Examples) taken from a transaction database.

➢ Training Samples: [Described through attribute values along with the class they belong to]

ID	Age	Income	Student	Credit Rating	*Decision/ Class/ Label*
1	≤ 30	high	no	fair	negative
2	≤ 30	high	no	excellent	negative
3	31…40	high	no	fair	positive
4	> 40	medium	no	fair	positive
5	> 40	low	yes	fair	positive
6	> 40	low	yes	excellent	negative
7	31…40	low	yes	excellent	positive
8	≤ 30	medium	no	fair	negative
9	≤ 30	low	yes	fair	positive
10	> 40	medium	yes	fair	positive
11	≤ 30	medium	yes	excellent	positive
12	31…40	medium	no	excellent	positive
13	31…40	high	yes	fair	positive
14	> 40	medium	no	excellent	negative

[See 'Data Mining: Concepts and Techniques, 3rd Edition, Morgan Kaufmann Publishers, 2012, by Jiawei Han, Micheline Kamber & Jian Pei]

- **3 parameters are computed:**

a) **Information Content** (I) regarding the classes in the samples computed as Entropy

$I(P(C_1), P(C_2), \ldots, P(C_m))$

$= \sum_{i=1:m} -P(C_i) \times Log_2(P(C_i))$,

where

$P(C_i)$ – probability of the class C_i ,

m – number of classes; Log_2 – binary encoding in information theory

For our example,

I(P(positive), P(negative))

$= -P(positive) \times Log_2(P(positive)) - P(negative) \times Log_2(P(negative))$

That is, $I(9/14, 5/14) = - 9/14 \times Log_2(9/14) - 5/14 \times Log_2(5/14)$

≈ 0.940 (bits)

[Only nontrivial probabilities (> 0 & <1) have been considered.]

b) **Remainder or expected entropy remaining after testing an attribute** A

Remainder(A) =

$\sum_{i=1:v}$ P(Samples with ith value of A) × I(Classes in samples with ith value of A),

where v = number of distinct values of A.

For our example,

i) Remainder(Age) = 5/14 × I(2/5, 3/5) [≤ 30]
+ 4/14 × I(4/4, 0/4) [31 … 40]
+ 5/14 × I(3/5, 2/5) [> 40]

$= 5/14 \times I(2/5, 3/5) + 5/14 \times I(3/5, 2/5)$
$= 2 \times 5/14 \times I(2/5, 3/5)$
$= 5/7 (- 2/5 Log_2(2/5) - 3/5 Log_2(3/5))$
≈ 0.694 (bits)

ii) Remainder(Income) = ?
iii) Remainder(Student) = ?
iv) Remainder(Credit Rating) = ?

[Self study]

c) **Information Gain** of an attribute test

Gain(A) = I(Classes in all samples in the table) – Remainder(A)

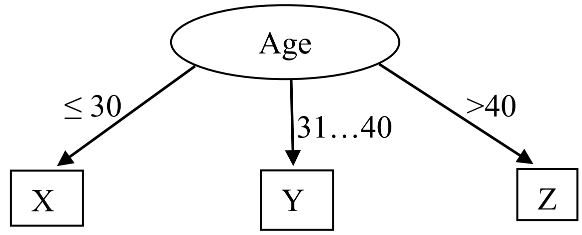
For our example,

Gain(Age) = 0.940 – 0.694 = 0.246 (bits)

** Check for yourselves that,

Gain(Income) = 0.029 bits
Gain(Student) = 0.151 bits
Gain(Credit Rating) = 0.048 bits

So, we have the attribute 'Age' with the highest Gain, and this leads to what follows.



X =

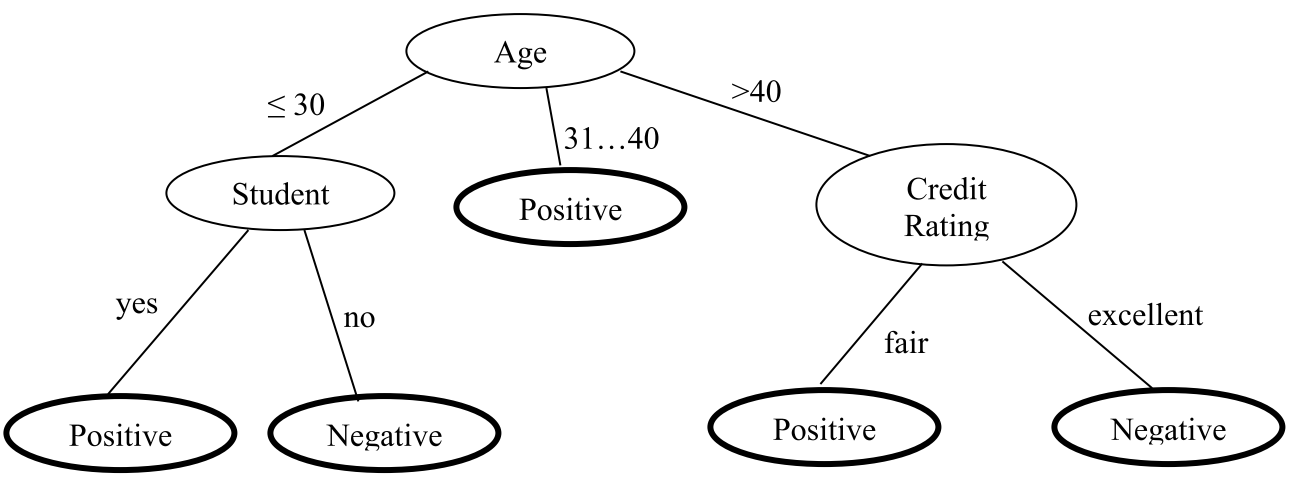
Age	**Income**	**Student**	**Credit Rating**	*Decision/ Class/ Label*
≤ 30	high	no	fair	negative
≤ 30	high	no	excellent	negative
≤ 30	medium	no	fair	negative
≤ 30	low	yes	fair	positive
≤ 30	medium	yes	excellent	positive

Y =

Age	**Income**	**Student**	**Credit Rating**	*Decision/ Class/ Label*
31…40	high	no	fair	positive
31…40	low	yes	excellent	positive
31…40	medium	no	excellent	positive
31…40	high	yes	fair	positive

Z = ? [Self study]

Finally we have the following tree. [Self study]



And it means that we have **learned the following 5 rules**.

1. If 'Age' = '≤ 30' and 'Student' = 'yes', then 'Class' = 'Buys a computer'.
2. If 'Age' = '≤ 30' and 'Student' = 'no', then 'Class' = 'Does not buy a computer'.
3. …..
4. …..
5. If 'Age' = '>40' and 'Credit Rating' = 'excellent', then 'Class' = 'Does not buy a computer'.